

WEATHER ELEMENTS / INDIA JET STREAM-WESTERN DISTURBANCES

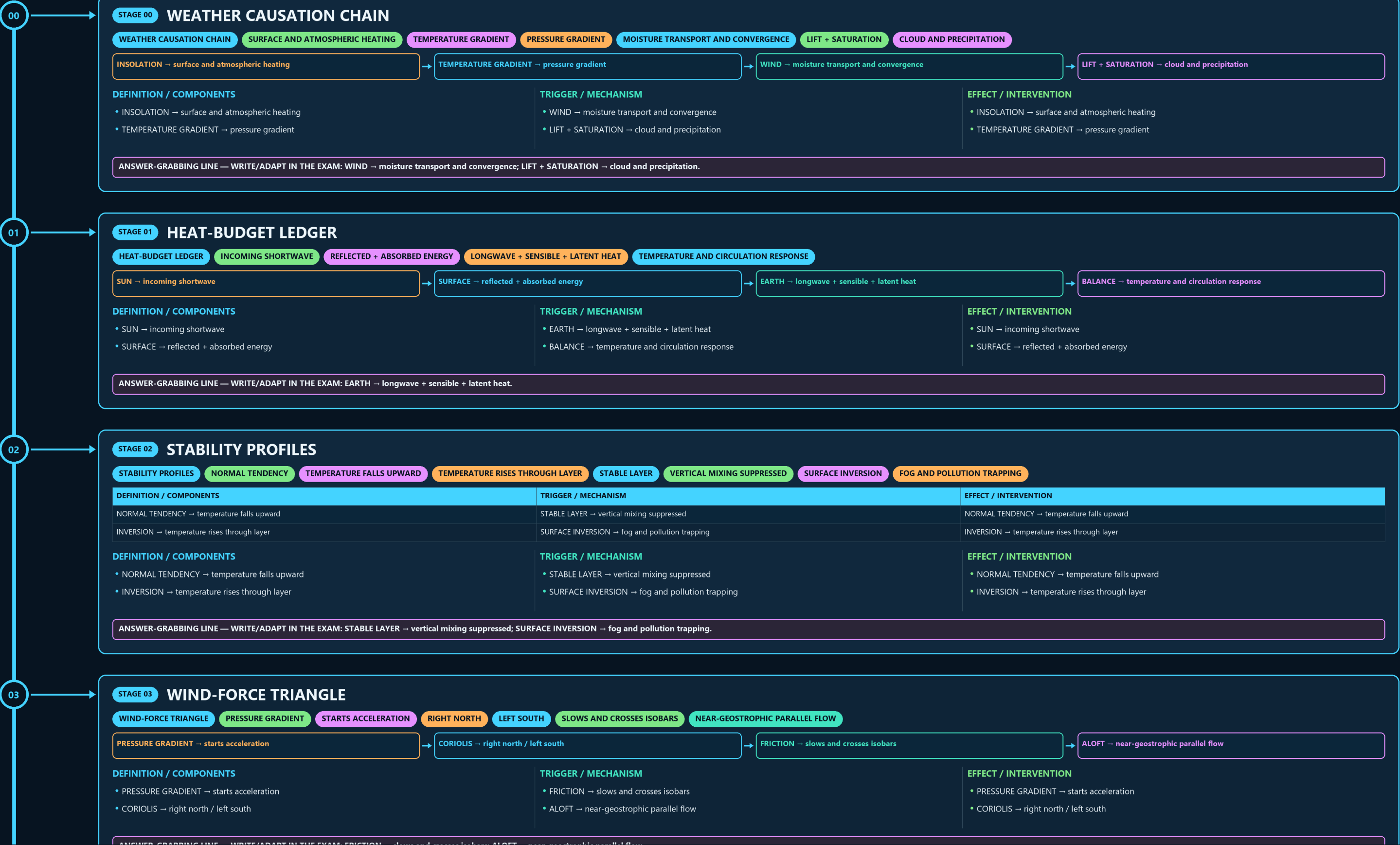
WEATHER CAUSATION CHAIN → HEAT-BUDGET LEDGER → STABILITY PROFILES → WIND-FORCE TRIANGLE → PLANETARY CIRCULATION RAIL → HUMIDITY-TO-CLOUD CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

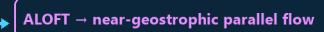


- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE LAYER → vertical mixing suppressed; SURFACE INVERSION → fog and pollution trapping.

STAGE 03 WIND-FORCE TRIANGLE

NEAR-GEOSTROPHIC PARALLEL FLOW



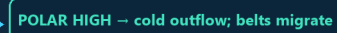
EFFECT / INTERVENTION

- PRESSURE GRADIENT → starts acceleration
- CORIOLIS → right north / left south

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRICTION → slows and crosses isobars; ALOFT → near-geostrophic parallel flow.

STAGE 04 PLANETARY CIRCULATION RAIL

POLAR HIGH



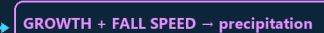
HAZARD / LIMIT / RESPONSE

- SUBPOLAR LOW → westerly-polar interaction
- POLAR HIGH → cold outflow; belts migrate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUBPOLAR LOW → westerly-polar interaction; POLAR HIGH → cold outflow; belts migrate.

STAGE 05 HUMIDITY-TO-CLOUD CHAIN

GROWTH + FALL SPEED



EFFECT / INTERVENTION

- MOISTURE CONTENT + TEMPERATURE → relative humidity
- COOLING TO SATURATION → dew point reached

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUCLEI + LIFT → droplets or ice crystals.

STAGE 06 RAINFALL COMPARISON

MECHANISMS MAY COMBINE

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
CONVECTIONAL → buoyant thermal ascent	FRONTAL / CYCLONIC → convergent dynamic ascent	CONVECTIONAL → buoyant thermal ascent
OROGRAPHIC → forced relief ascent	REAL EVENT → mechanisms may combine	OROGRAPHIC → forced relief ascent
DEFINITION / COMPONENTS • CONVECTIONAL → buoyant thermal ascent • OROGRAPHIC → forced relief ascent	TRIGGER / MECHANISM • FRONTAL / CYCLONIC → convergent dynamic ascent • REAL EVENT → mechanisms may combine	EFFECT / INTERVENTION • CONVECTIONAL → buoyant thermal ascent • OROGRAPHIC → forced relief ascent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRONTAL / CYCLONIC → convergent dynamic ascent; REAL EVENT → mechanisms may combine.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUCLEI + LIFT → droplets or ice crystals.

06

STAGE 06

RAINFALL COMPARISON

RAINFALL COMPARISON

BUOYANT THERMAL ASCENT

FORCED RELIEF ASCENT

CONVERGENT DYNAMIC ASCENT

REAL EVENT

MECHANISMS MAY COMBINE

DEFINITION / COMPONENTS

CONVECTIONAL → buoyant thermal ascent

OROGRAPHIC → forced relief ascent

DEFINITION / COMPONENTS

• CONVECTIONAL → buoyant thermal ascent

• OROGRAPHIC → forced relief ascent

TRIGGER / MECHANISM

FRONTAL / CYCLONIC → convergent dynamic ascent

REAL EVENT → mechanisms may combine

TRIGGER / MECHANISM

• FRONTAL / CYCLONIC → convergent dynamic ascent

• REAL EVENT → mechanisms may combine

EFFECT / INTERVENTION

CONVECTIONAL → buoyant thermal ascent

OROGRAPHIC → forced relief ascent

EFFECT / INTERVENTION

• CONVECTIONAL → buoyant thermal ascent

• OROGRAPHIC → forced relief ascent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRONTAL / CYCLONIC → convergent dynamic ascent; REAL EVENT → mechanisms may combine.

07

STAGE 07

TROPOSPHERE SECTION

TROPOSPHERE SECTION

MOISTURE, FRICTION AND HEATING

MID LEVELS

CLOUDS, TROUGHS AND STABILITY

UPPER TROPOSPHERE

JETS AND DIVERGENCE

TROPICS DEEPER

POLES SHALLOWER

SURFACE → moisture, friction and heating

MID LEVELS → clouds, troughs and stability

UPPER TROPOSPHERE → jets and divergence

TROPICS DEEPER / POLES SHALLOWER

SYSTEM / DEFINITION

• SURFACE → moisture, friction and heating

• MID LEVELS → clouds, troughs and stability

PROCESS / MECHANISM

• UPPER TROPOSPHERE → jets and divergence

• TROPICS DEEPER / POLES SHALLOWER

SPATIAL / INDIA PATTERN

• SURFACE → moisture, friction and heating

• MID LEVELS → clouds, troughs and stability

HAZARD / LIMIT / RESPONSE

• UPPER TROPOSPHERE → jets and divergence

• TROPICS DEEPER / POLES SHALLOWER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UPPER TROPOSPHERE → jets and divergence; TROPICS DEEPER / POLES SHALLOWER.

08

STAGE 08

JET-STREAM ENGINE

JET-STREAM ENGINE

HORIZONTAL TEMPERATURE GRADIENT

PRESSURE GRADIENT ALOFT

CORIOLIS BALANCE

STRONG NEAR-ZONAL WIND

ROSSBY WAVES

RIDGES, TROUGHS AND STEERING

HORIZONTAL TEMPERATURE GRADIENT

HEIGHT / PRESSURE GRADIENT ALOFT

CORIOLIS BALANCE → strong near-zonal wind

ROSSBY WAVES → ridges, troughs and steering

DEFINITION / COMPONENTS

• HORIZONTAL TEMPERATURE GRADIENT

• HEIGHT / PRESSURE GRADIENT ALOFT

TRIGGER / MECHANISM

• CORIOLIS BALANCE → strong near-zonal wind

• ROSSBY WAVES → ridges, troughs and steering

EFFECT / INTERVENTION

• HORIZONTAL TEMPERATURE GRADIENT

• HEIGHT / PRESSURE GRADIENT ALOFT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CORIOLIS BALANCE → strong near-zonal wind; ROSSBY WAVES → ridges, troughs and steering.

09

STAGE 09

INDIA SEASONAL JETS

INDIA SEASONAL JETS

STWJ SOUTH OF HIMALAYA SECTOR

STEERS WESTERN DISTURBANCES EAST

TROPICAL EASTERLY JET DEVELOPS

JETS ARE PARTS OF WIDER CIRCULATION

WINTER → STWJ south of Himalaya sector

STWJ → steers western disturbances east

SUMMER → tropical easterly jet develops

RULE → jets are parts of wider circulation

SYSTEM / DEFINITION

• WINTER → STWJ south of Himalaya sector

• STWJ → steers western disturbances east

PROCESS / MECHANISM

• SUMMER → tropical easterly jet develops

• RULE → jets are parts of wider circulation

SPATIAL / INDIA PATTERN

• WINTER → STWJ south of Himalaya sector

• STWJ → steers western disturbances east

HAZARD / LIMIT / RESPONSE

• SUMMER → tropical easterly jet develops

• RULE → jets are parts of wider circulation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → jets are parts of wider circulation.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CORIOLIS BALANCE → strong near-zonal wind; ROSSBY WAVES → ridges, troughs and steering.

09

STAGE 09 INDIA SEASONAL JETS

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SPATIAL / INDIA PATTERN

- WINTER → STWJ south of Himalaya sector
- STWJ → steers western disturbances east

HAZARD / LIMIT / RESPONSE

- SUMMER → tropical easterly jet develops
- RULE → jets are parts of wider circulation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → jets are parts of wider circulation.

10

STAGE 10 WESTERN-DISTURBANCE CHAIN

WESTERN-DISTURBANCE CHAIN

UPPER TROUGH

EASTWARD WESTERLY MOTION

MOISTURE + DYNAMICAL LIFT

CLOUD AND RAIN/SNOW

HIMALAYA + PLAINS

TERRAIN-SPECIFIC IMPACTS

BENEFICIAL OR DAMAGING BY TIMING/INTENSITY

UPPER TROUGH / LOW → eastward westerly motion

MOISTURE + DYNAMICAL LIFT → cloud and rain/snow

HIMALAYA + PLAINS → terrain-specific impacts

OUTCOME → beneficial or damaging by timing/intensity

DEFINITION / COMPONENTS

- UPPER TROUGH / LOW → eastward westerly motion
- MOISTURE + DYNAMICAL LIFT → cloud and rain/snow

TRIGGER / MECHANISM

- HIMALAYA + PLAINS → terrain-specific impacts
- OUTCOME → beneficial or damaging by timing/intensity

EFFECT / INTERVENTION

- UPPER TROUGH / LOW → eastward westerly motion
- MOISTURE + DYNAMICAL LIFT → cloud and rain/snow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MOISTURE + DYNAMICAL LIFT → cloud and rain/snow.

11

STAGE 11 FORECAST EVIDENCE LADDER

FORECAST EVIDENCE LADDER

PILOT BALLOON

UPPER-AIR STATE

SATELLITE + RADAR

CLOUD AND PRECIPITATION

NWP ENSEMBLE

SCENARIOS AND UNCERTAINTY

ACTION; LATER OBSERVATIONS VERIFY OUTCOME

RADIOSONDE / PILOT BALLOON → upper-air state

SATELLITE + RADAR → cloud and precipitation

NWP ENSEMBLE → scenarios and uncertainty

WARNING → action; later observations verify outcome

DEFINITION / COMPONENTS

- RADIOSONDE / PILOT BALLOON → upper-air state
- SATELLITE + RADAR → cloud and precipitation

TRIGGER / MECHANISM

- NWP ENSEMBLE → scenarios and uncertainty
- WARNING → action; later observations verify outcome

EFFECT / INTERVENTION

- RADIOSONDE / PILOT BALLOON → upper-air state
- SATELLITE + RADAR → cloud and precipitation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NWP ENSEMBLE → scenarios and uncertainty; WARNING → action; later observations verify outcome.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

JET STREAMS ARE FAST, NARROW UPPER-TROPOSPHERIC WIND BANDS.

STWJ LIES SOUTH OF THE HIMALAYAS IN WINTER AND

NORTHWARD WITHDRAWAL OF STWJ IS A PRE-CONDITION FOR SW

TEJ DEVELOPS AS PART OF THE SUMMER MONSOON'S UPPER-AIR

WESTERN DISTURBANCES ARE EXTRA-TROPICAL, NON-MONSOONAL LOW-PRESSURE SYSTEMS FROM THE

WDS BRING WINTER/EARLY-SPRING RAIN TO PUNJAB, HARYANA, DELHI AND

THEY ARE VITAL FOR RABI CROPS SUCH AS WHEAT

OPTIONAL PROCESS DEPTH

- Jet streams are fast, narrow upper-tropospheric wind bands.
- STWJ lies south of the Himalayas in winter and acts as the conveyor for Western Disturbances.
- Northward withdrawal of STWJ is a pre-condition for SW monsoon onset.
- TEJ develops as part of the summer monsoon's upper-air circulation. Tibetan/Asian heating matters,

DATA / INSTITUTION LIMIT

- Western Disturbances are extra-tropical, non-monsoonal low-pressure systems from the Mediterranean moving east via Iran-Afghanistan-Pakistan.
- WDs bring winter/early-spring rain to Punjab, Haryana, Delhi and UP and snow to J&K, HP and Uttarakhand
- they are vital for Rabi crops such as wheat, mustard and gram.
- In May–June, intense heating over north-west India supports the hot, dry Loo. Its timing and strength

CONTEMPORARY PARALLEL

- Altitude lowers temperature by about 6.5 deg C/km in the notes
- the Himalayan barrier gives India warmer winters than comparable latitudes by blocking Central Asian cold air.
- STWJ steers Western Disturbances in winter.
- STWJ's northward shift is essential for SW monsoon onset.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.